

\$577.60K

Total Revenue

46K

Units Sold

150.63

Average Price

38.21

Revenue per Unit

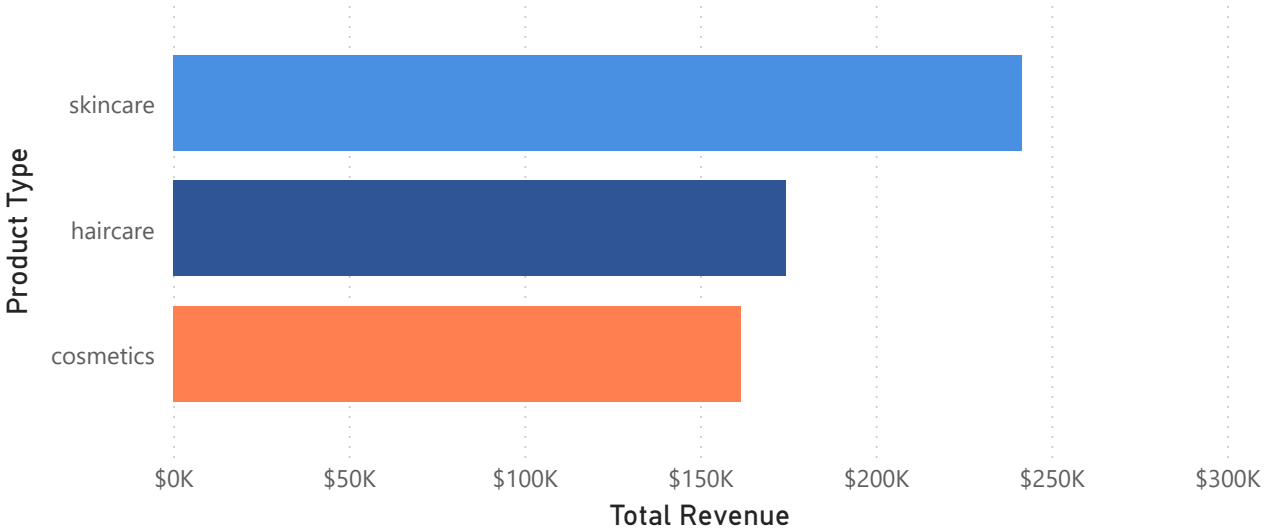
17.38K

Revenue per SKU

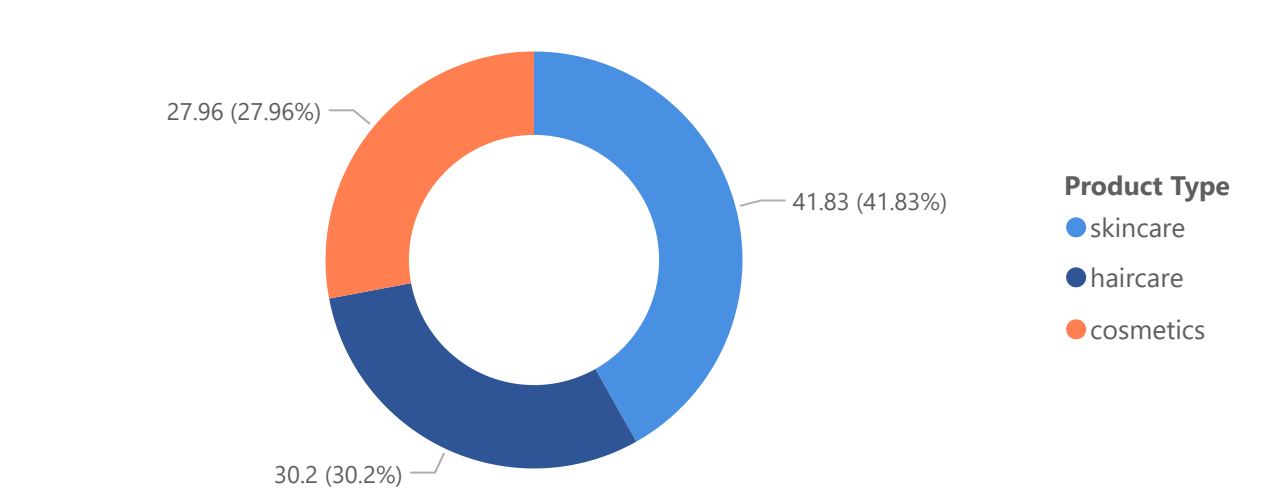
skincare

Top Product Type

Total Revenue by Product Type



Percent of Total Revenue by Product Type



product type	Total Revenue	Units Sold	Revenue per Unit	Revenue per SKU	Percent of total revenue	Average Price
cosmetics	161521.27	11757	13.74	6212.36	28	57.36
skincare	241628.16	20731	11.66	6040.70	42	47.26
haircare	174455.39	13611	12.82	5131.04	30	46.01
Total	577604.82	46099	38.21	17384.10	100	150.63

- Key Insights:
- Skincare drives 41.8% of total revenue, outperforming haircare and cosmetics.
 - Cosmetics has highest average price but lowest total revenue, indicating pricing sensitivity.
 - Haircare is middle in both units and revenue, showing stable category performance.

- Recommendations:
- Expand skincare SKUs to increase share in top-performing category.
 - Review pricing elasticity on cosmetics to capture more volume.
 - Maintain haircare production as a stable baseline contributor.

100

Total Records

36

Fail Count

36%

Overall Fail Rate %

\$1.70K

Estimated Defect Cost

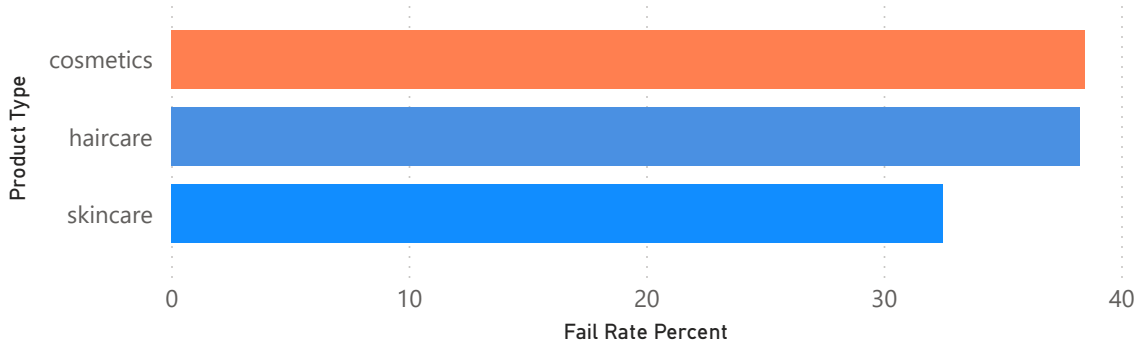
2.24

Average Defect Rate

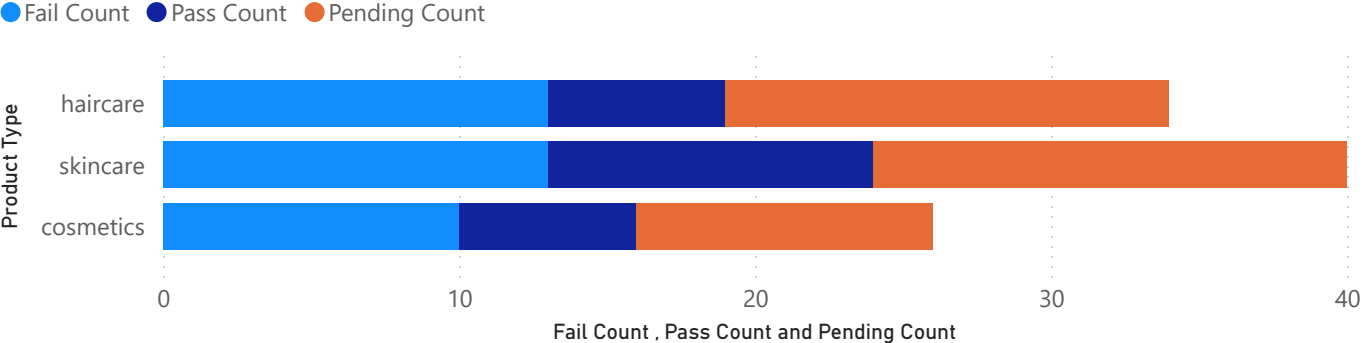
haircare

Highest Risk Product

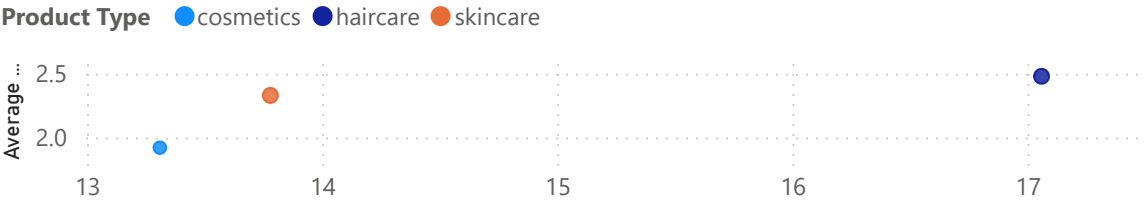
Fail Rate Percent by Product Type



Fail Count , Pass Count and Pending Count by Product Type



Average Manufacturing Lead Time , Average Defect Rate and Estimated Defect Cost by Product Type



Key Insights

- Haircare is the highest-risk product type, driven by the strongest combination of high fail rate (38.24%) and elevated average defect rate.
- Cosmetics has the highest fail rate at 38.46%, indicating immediate quality control attention, but its defect cost impact is lower than haircare due to fewer units produced.
- Skincare shows the lowest fail rate (32.5%) and lowest average defect rate, making it the most stable category operationally.
- Pending inspections are weighted more toward haircare, indicating potential hidden failure volume not yet reflected in totals.
- Lead time variation across product types suggests inconsistencies in manufacturing cycle control and supplier performance.

Product Type	Total Records	Fail Count	Pass Count	Fail Rate Percent	Average Defect Ra
cosmetics	26	10	6	38.46	1
haircare	34	13	6	38.24	2
skincare	40	13	11	32.5	2
Total	100	36	23		2.

Recommendations

- Prioritize root-cause corrective action on haircare defects to mitigate future escalations and reduce its risk contribution.
- Conduct supplier process reviews for cosmetics, focusing on tightening defect thresholds and QC acceptance criteria to reduce fail volume.
- Standardize manufacturing checkpoints for skincare, as it shows best-in-class performance and can act as a benchmark process for the other two categories.
- Implement in-line quality screening earlier in the manufacturing cycle to reduce downstream defect cost exposure.
- Introduce supplier quality scorecards to improve accountability, especially in categories with longer lead times and higher defect tendencies.

14.72

Average Manufacturing Lead...

15.87

Average Supplier Lead Time

1.16

Supply Chain Gap (Days)

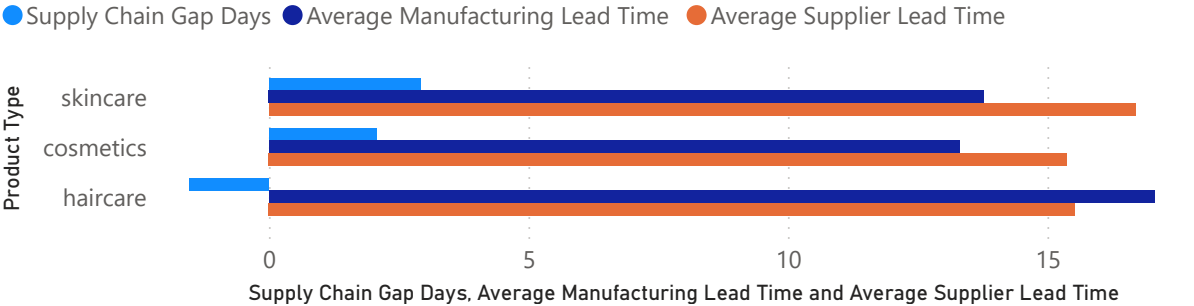
cosmetics

Highest Lead Time Variability

skincare

Most Delayed Product

Supply Chain Gap Days, Average Manufacturing Lead Time and Average Supplier Lead Time by Product Type

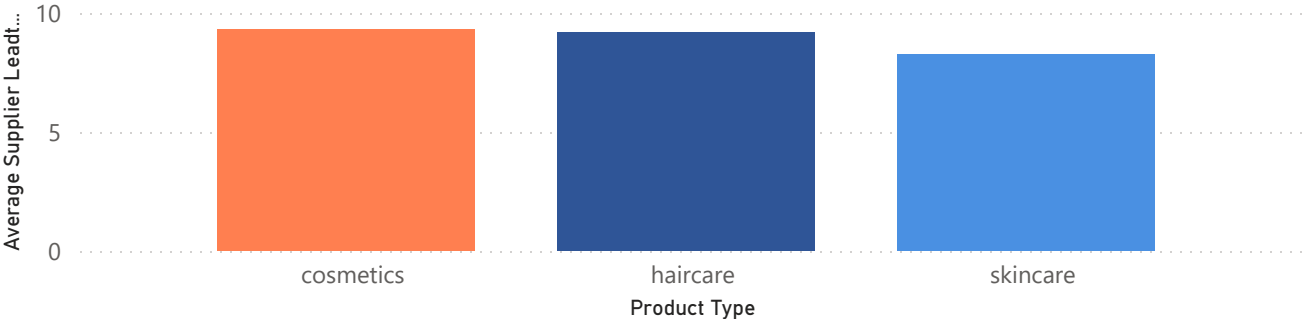


Product Type	Average Manufacturing Lead Time	Average Supplier Lead Time	Supply Chain Gap Day
cosmetics	13.31	15.38	2.0
haircare	17.06	15.53	-1.5
skincare	13.78	16.70	2.9

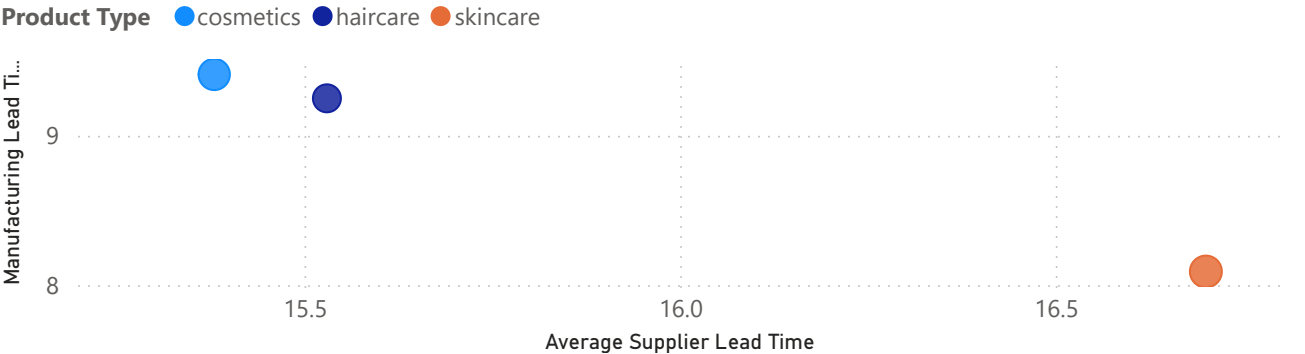
Key Insights

- Manufacturing lead times are generally stable across product types, while supplier lead times create the primary variability in full cycle timing.
- The gap between manufacturing lead time and supplier lead time indicates that upstream scheduling and inbound logistics, not production flow, are the main contributors to delay risk.
- Product categories with wider supplier lead time variation represent a higher likelihood of stockout, expediting fees, and postponed release dates.

Average Supplier Leadtime Variance by Product Type



Sum of Lead Time Ratio by Product Type , Average Supplier Lead Time and Manufacturing Lead Time Variance



Recommendations

- Implement supplier timing controls and expectations through updated SLAs that attach measurable consequence to late or variable delivery windows.
- Introduce mid-cycle supplier check-ins to confirm shipment readiness before manufacturing completes, reducing idle storage, backlog, and rescheduling.
- Prioritize lead time stabilization with the highest-variance supplier categories first, rather than focusing on mean duration alone.

\$10.81K

Total Expected Quality

\$1.70K

Estimated Defect Cost

\$46.83

Average Manufacturing Cost

\$105.78

Expected Quality Cost Per Unit

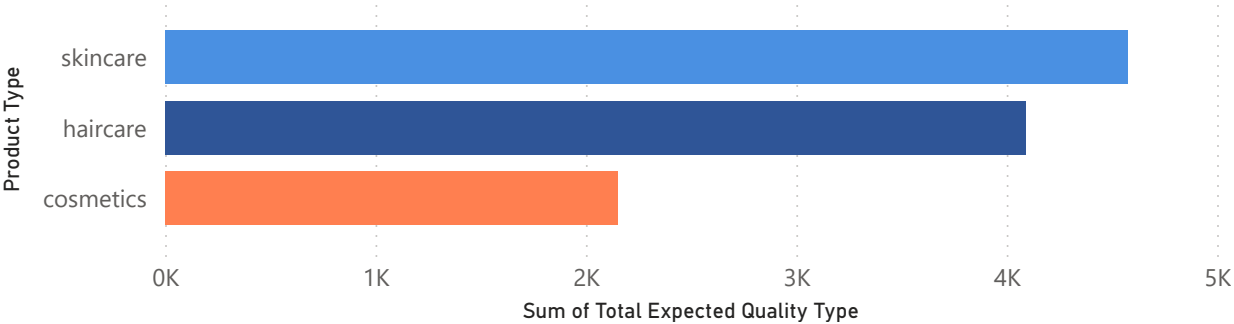
skincare

Highest Cost...

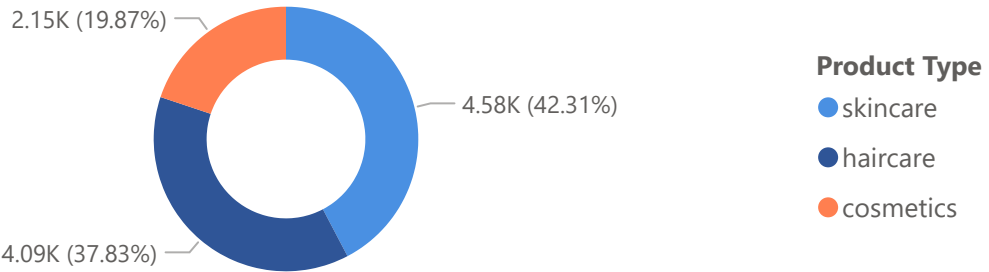
cosmet...

Lowest Cost...

Sum of Total Expected Quality Type by Product Type



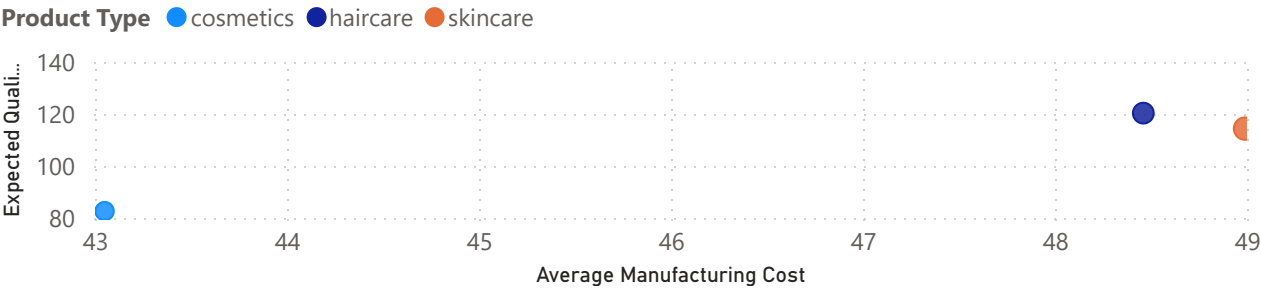
Sum of Total Expected Quality Cost by Product Type



Key Financial Insights

- Skincare carries the highest share of total defect-related cost exposure, representing approximately 42 percent of all quality cost impact.
- Haircare follows at nearly 38 percent, indicating consistent quality cost leakage that is tied more to defect frequency than to base manufacturing expense.
- Cosmetics shows the lowest total cost exposure at around 20 percent, but this is due to lower unit volume, not a lower quality risk profile. If volume increases, associated cost exposure will scale rapidly.

Total Units by Product Type, Average Manufacturing Cost and Expected Quality Cost Per Unit



Product Type	Total Units	Average Manufacturing Cost	Average Defect Rate	Estimated Defect Cost	Ex
cosmetics	26	43.05	1.92	430.53	
haircare	34	48.46	2.48	629.95	
skincare	40	48.99	2.33	636.91	

Key Financial Insights

- Focus defect reduction efforts on skincare first, as any reduction in failure rates will yield the most immediate financial improvement due to its cost concentration.
- Stabilize haircare quality processes through tighter tolerance checks, defect pattern reviews, and mid-line inspection points to prevent recurring rework accumulation
- Avoid scaling cosmetics production until quality controls are reinforced, since current low exposure is volume-dependent rather than performance-based.